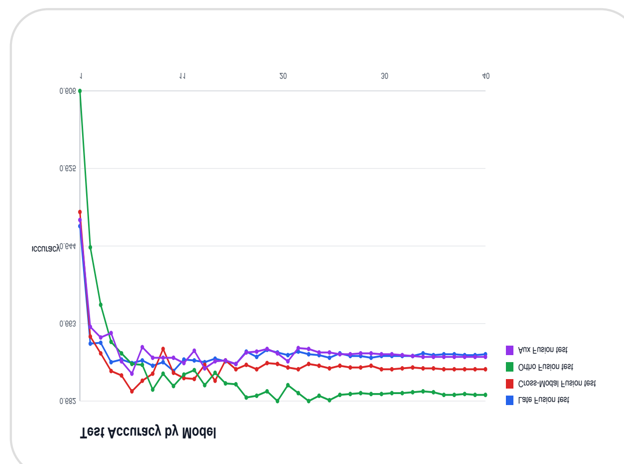
Overview

M3Sentiment is a transformer-based multimodal sentiment analysis project on CMU-MOSEI. It predicts whether a video clip is negative, neutral, or positive by combining text, audio, and visual signals.

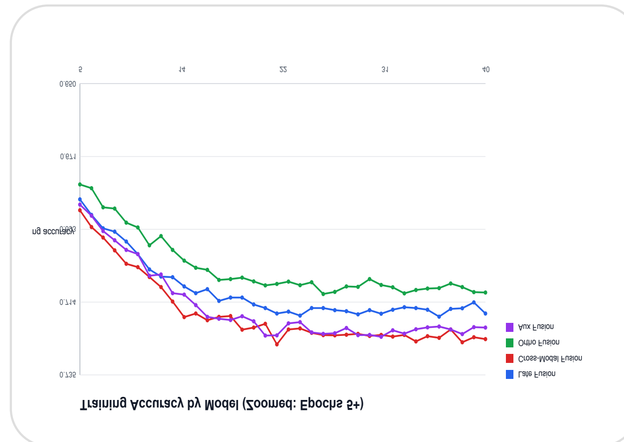
- Main task: 3-way sentiment classification from aligned text, audio, and vision sequences.
- Core comparison: Late Fusion, Cross-Modal Attention, Cross-Modal + Orthogonality Loss, and Cross-Modal + Auxiliary Heads.
- Main contribution: beyond accuracy, the project tracks internal diagnostics such as modality attention, orthogonality, attention entropy, and loss components.

Text | Audio | Vision

Final takeaway: smarter fusion helps modestly, while diagnostics explain why models behave differently.



Final test accuracy by model



Training accuracy over epochs

Data And Preprocessing

The project expects a processed CMU-MOSEI dataset at data/aligned_mosei_dataset.pkl. Each video clip is represented as three aligned sequences with length 50: text, audio, and vision.

Component	Implementation
Dataset loader	Loads train/valid/test splits from aligned_mosei_dataset.pkl
Normalization	Computes mean/std on train split and reuses them for validation/test
Audio cleanup	Replaces NaN and ±inf audio values with zero
Labels	Uses 3-class labels: negative, neutral, positive
Dataloaders	Train loader shuffles; validation/test loaders are deterministic

Implementation files: src/m3sentiment/dataset.py and src/m3sentiment/data_loaders.py. Data utilities for download, CMU Multimodal SDK processing, exporting, and inspection are under data/scripts/.

Data flow

```
CMU-MOSEI video clip → aligned text/audio/vision features → normalized tensors → transformer model
```

Model Architectures

All models share a common transformer-based multimodal backbone. Each modality is projected into a shared hidden dimension, receives positional encoding, passes through transformer encoder layers, and is pooled into one summary vector.

Model	What it implements	Why it matters
Late Fusion	Separate modality encoders plus final fusion transformer	Simple multimodal baseline
Cross-Modal	Each modality attends to the other two modalities	Models disagreement and interaction
Ortho Fusion	Cross-modal model plus orthogonality loss	Reduces redundant modality features
Aux Fusion	Cross-modal model plus text/audio/vision auxiliary heads	Forces each branch to learn sentiment cues

Core pipeline

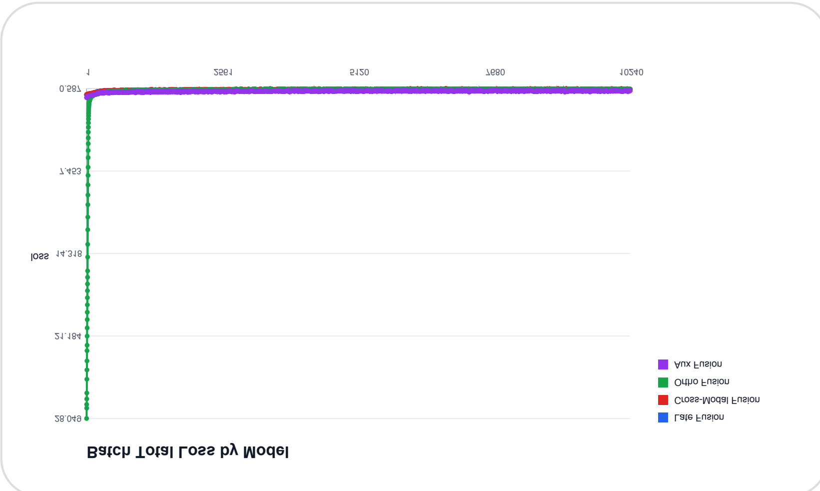
```
Text sequence   → Text Transformer   → Text vector
Audio sequence  → Audio Transformer  → Audio vector
Vision sequence → Vision Transformer → Vision vector
```

The architecture uses instrumented transformer layers, so attention maps are available for later diagnostic analysis.

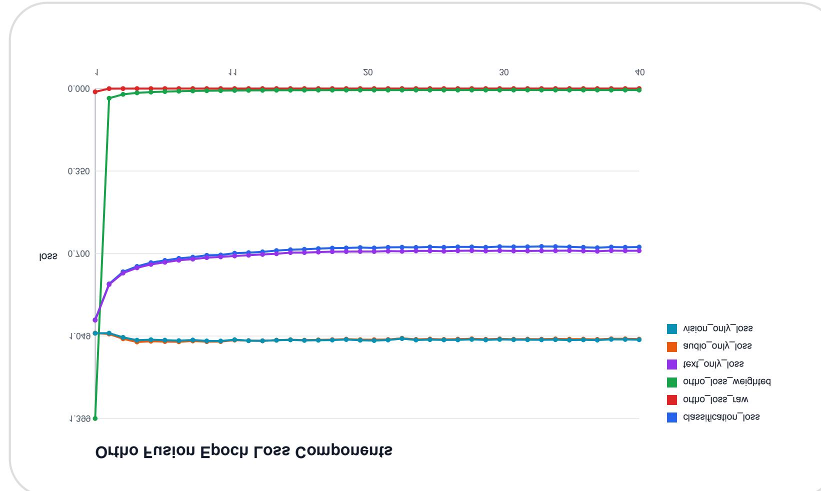
Training And Loss Functions

Training is implemented in scripts/train_models.py and src/m3sentiment/training.py. All models use CrossEntropyLoss, Adam, gradient clipping, and ReduceLROnPlateau scheduling.

Model	Training objective
Late Fusion	`classification loss`
Cross-Modal	`classification loss`
Ortho Fusion	`classification loss + 100 × orthogonality loss`
Aux Fusion	`main fused loss + 0.05 × (text aux + audio aux + vision aux)`



Batch total loss comparison

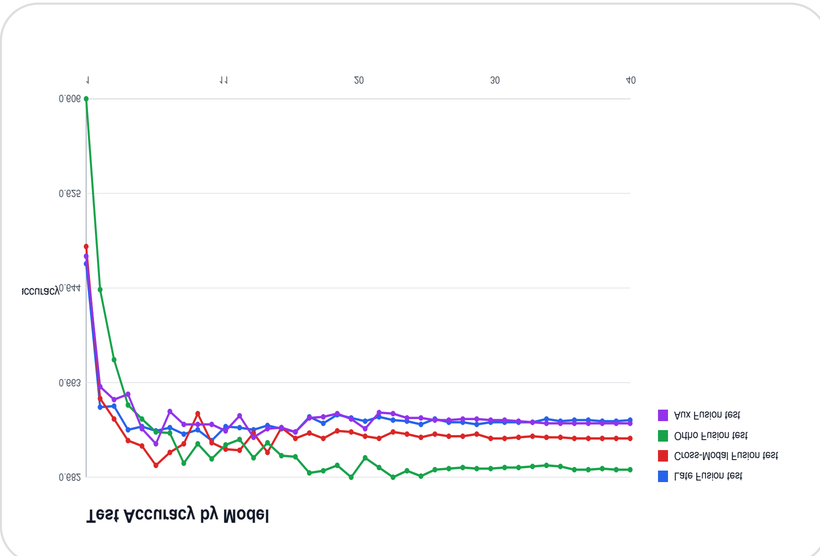


Ortho loss components

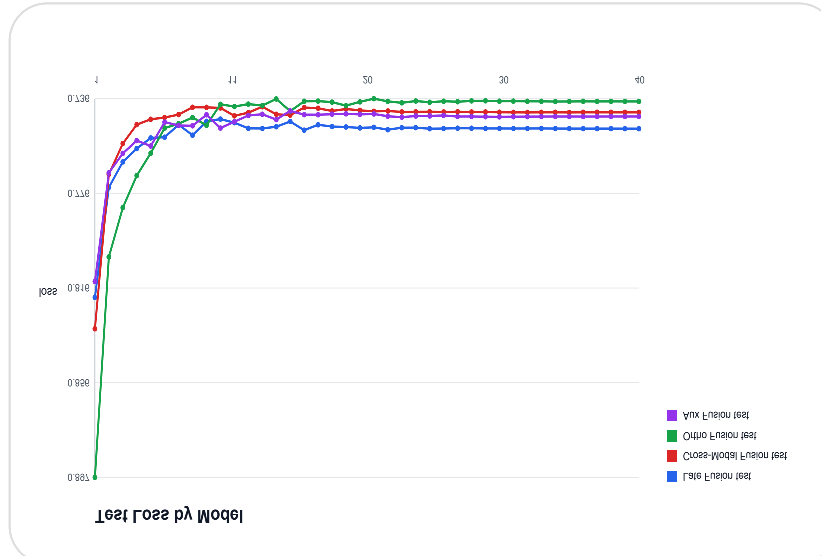
Final config: batch size 64, learning rate 2e-5, 40 epochs, hidden dimension 128, 4 attention heads, 2 transformer layers, dropout 0.1.

Evaluation Results

Model	Final Test Acc	Best Test Acc	Macro F1	Weighted F1
Late Fusion	67.03%	67.44%	61.22%	65.66%
Cross-Modal	67.40%	67.93%	61.43%	65.90%
Ortho Fusion	68.02%	68.17%	61.79%	66.41%
Aux Fusion	67.10%	67.50%	61.14%	65.63%



Final test accuracy



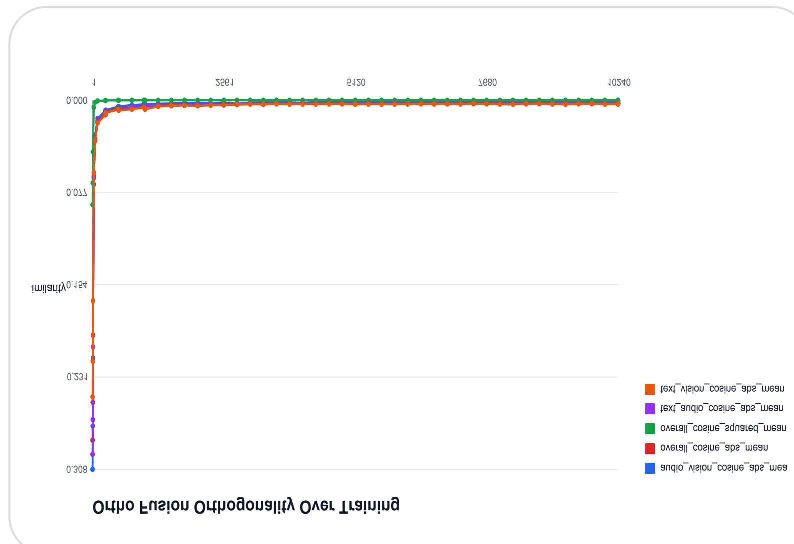
Final test loss

Key result: Cross-Modal + Orthogonality achieves the best final accuracy and best F1 values in the finalized run. This suggests that reducing modality redundancy can improve generalization.

Representation Diagnostics

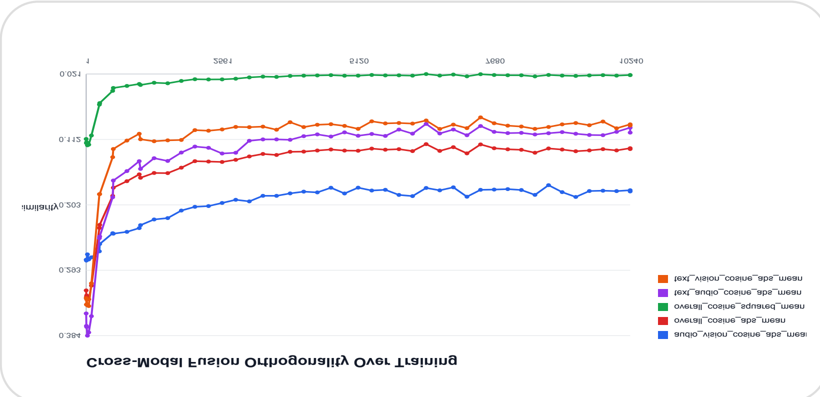
Orthogonality diagnostics measure how similar the learned text, audio, and vision vectors are. Lower similarity means the modalities are learning more distinct information.

Model	Avg Modality Similarity
Late Fusion	0.2529
Cross-Modal	0.1251
Ortho Fusion	0.0025
Aux Fusion	0.1759

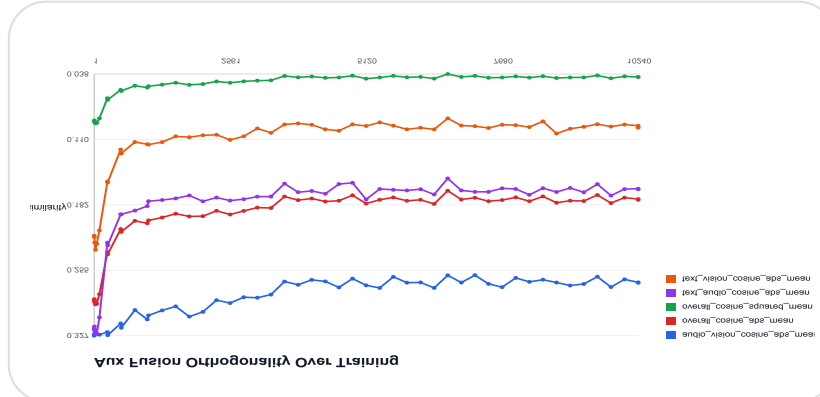


Orthogonality over training

The Orthogonality model successfully drives modality similarity close to zero. This is the clearest evidence that the added loss changed what the model learned internally.



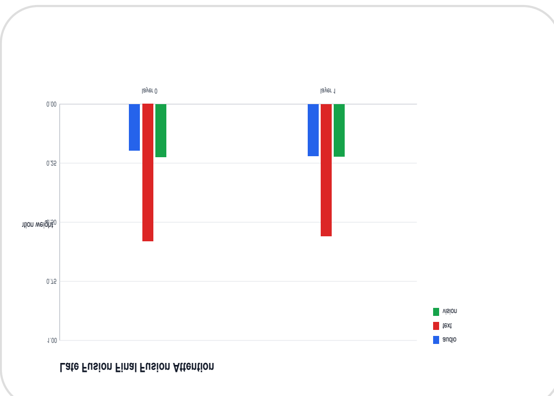
Cross-Modal similarity trend



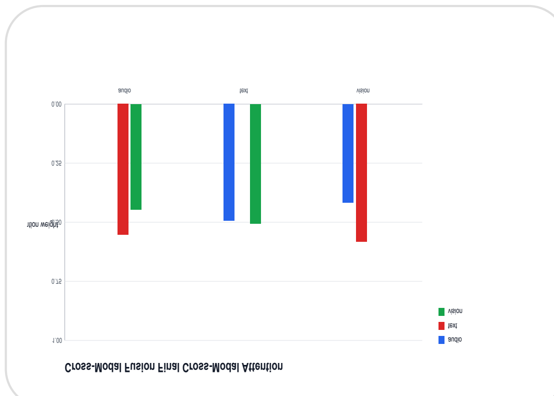
Aux Fusion similarity trend

Attention Diagnostics

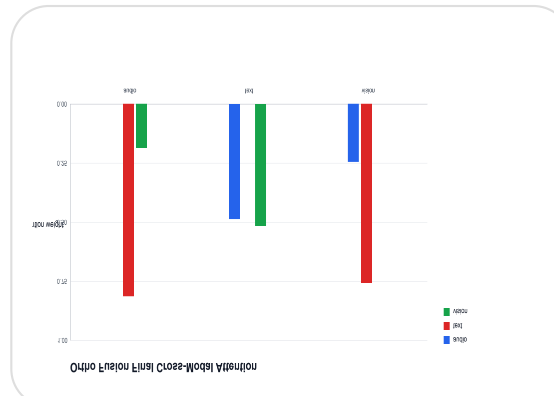
The diagnostic framework tracks how much attention is paid to each modality and how attention changes across heads, layers, batches, and epochs.



Late Fusion

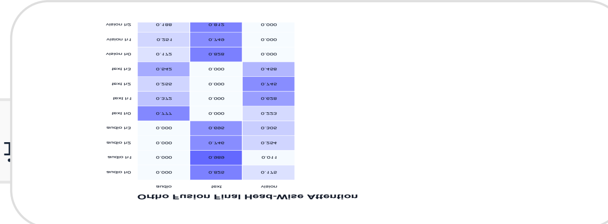


Cross-Modal



Ortho Fusion

Observation	Interpretation
Late Fusion attends most to text	Text is the dominant sentiment signal
Cross-Modal attention is more balanced	Modalities interact more directly
Ortho audio/vision attend strongly to text	Text becomes the anchor after modality separation
Head-level plots expose variation	Different attention heads can speci



Head-level attention heatmap

Failure Analysis

Confusion matrices show that all models are much better at detecting positive and negative sentiment than neutral sentiment.

Model	Negative Recall	Neutral Recall	Positive Recall
Late Fusion	71.63%	31.12%	80.43%
Cross-Modal	72.74%	30.63%	80.74%
Ortho Fusion	72.67%	30.15%	82.27%
Aux Fusion	72.15%	30.54%	80.52%

Ortho Fusion confusion matrix

Cross-Modal confusion matrix

Main failure mode: neutral examples are often pushed into positive or negative classes because they contain weak or ambiguous emotional cues.

Conclusions And Future Work

- The project implements four transformer-based multimodal sentiment models and compares them under the same training setup.
- Cross-Modal + Orthogonality achieves the best final test accuracy in the finalized outputs.
- Orthogonality diagnostics confirm that the orthogonality loss almost completely separates text, audio, and vision representations.
- Attention diagnostics show that text remains a strong anchor, especially for audio and vision queries.
- Neutral sentiment is the hardest class and is the main opportunity for future improvement.

Future work

- Tune orthogonality and auxiliary loss weights.
- Use class-balanced loss or sampling to improve neutral sentiment.
- Select best-validation checkpoints instead of only final epoch weights.
- Run multiple random seeds for stronger statistical confidence.
- Try larger pretrained multimodal models and add qualitative example-level analysis.

Final takeaway: M3Sentiment shows that internal diagnostics are as important as accuracy for understanding multimodal